

Mapping WHO 2026 heat–health action-plan elements onto Hong Kong’s existing warning-and-shelter instruments

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2 September 2026

Running title. Hong Kong instruments and WHO 2026 HHAP elements

Provenance. Companion to the live identification article. It does not replace that article’s Table 2, freeze a confirmatory thermal claim, or evaluate Hong Kong warnings. Numerals 1.022 and 1.073 are quoted from the identification article as context only.

Abstract

Background. The World Health Organization Regional Office for Europe’s 2026 heat–health action-plan (HHAP) guidance restates eight core elements and eight corresponding outputs. Hong Kong already operates heat and cold warnings, a prolonged-heat alert that names hot nights, risk-group advice that names heart disease, and temporary shelters. Whether that instrument bundle constitutes a documented HHAP is a different question from whether a monthly cardiac-hospitalisation panel can evaluate it.

Methods. We mapped each 2026 core element onto named Hong Kong instruments using published WHO outputs and primary service pages. We then asked, for each element, whether a 132-month ecological panel of first coronary heart disease (CHD) and heart failure (HF) hospitalisations among people with type 2 diabetes and/or hypertension can evaluate the instrument. Admission cause was not recorded in that panel. Reported count ratios remain those of Model 1 in the companion identification article.

Results. Hong Kong has operational products for warnings, risk-group naming, public messaging, and temporary shelters. It has not, in the sources reviewed here, published the eight WHO *output documents* (an agreed HHAP governance structure, an alert-to-action package with assigned timelines, a mapped outreach strategy, a communication-strategy document, health-system business continuity plans, an exposure-reduction roadmap, operational heat–health surveillance, and a monitoring-evaluation-learning framework). The monthly panel cannot supply those missing outputs. It also cannot evaluate the products that do exist. A daily Hong Kong study of mortality and emergency calls during 2012–2022 is a different estimand and is not this panel.

Conclusions. The honest HHAP contribution of the present research programme is a map, not an evaluation. A heat-only reading would omit the more coherent residual in the companion article, which is HF with official cold days. Evaluation of warnings would require daily, cause-recorded outcomes, a predeclared contrast, and warning-day exposure. Those data are not this file.

Keywords: heat–health action plan; Hong Kong; Very Hot Weather Warning; Prolonged Heat Special Alert; surveillance; evaluation

Introduction

Heat–health action plans are a public-health response to extreme heat [1]. The first WHO/Europe guidance appeared in 2008. A 2019 commentary argued that, a decade later, evaluation, subnational delivery, and longer-term exposure reduction remained weak even where warning systems existed [2]. The 2026 second edition is implementation-oriented. It is organised around eight core elements and eight documented outputs [1]. It is written for the WHO European Region and is stated to be adaptable elsewhere. It does not prescribe a single national model.

Hong Kong is not a European member state. It is a dense subtropical city that already issues heat and cold warnings [3], a prolonged-heat alert that names hot nights [4,5], Centre for Health Protection advice that names heart disease or high blood pressure [6], and temporary heat and cold shelters [7]. The question for a supervisor who cares about HHAPs is not whether those pages exist. They do. The question is whether they match the 2026 outputs, and whether a monthly first-event panel can test them.

A companion identification article reports ecological monthly count ratios for first CHD and HF hospitalisation among people with type 2 diabetes and/or hypertension in Hong Kong, January 2013–December 2023

[8]. Under Newey–West lag-6 reporting, the count ratio was 1.022 (1.002–1.042) per five official hot nights for CHD and 1.073 (1.006–1.144) per five official cold days for HF. All twelve Model 1 Benjamini–Hochberg q -values exceeded 0.19. Those numerals are identification facts. They are not evidence that a warning works. This paper therefore does three things. It maps the eight 2026 elements onto named Hong Kong instruments. It records, for each element, whether the WHO *output document* is in hand. It states what the monthly panel cannot evaluate, including why a heat-only HHAP reading would miss the colder residual.

Methods

Mapping protocol

The unit of analysis is an instrument or document, not a person and not a hospital count. For each WHO 2026 core element we recorded: (i) the published output required by the guidance; (ii) the named Hong Kong product, if any; (iii) the primary URL or paper; (iv) whether that product is the WHO output or a related instrument; (v) what the 132-month CHD/HF panel can and cannot say.

Sources were the 2026 guidance and executive summary [1]; the Hong Kong Observatory service page for Cold and Very Hot Weather Warnings [3]; the 16 July 2022 Prolonged Heat service note [5]; Chong and Law’s December 2023 education page [4]; the Centre for Health Protection heat-stroke page dated 11 August 2025 [6]; and the Home Affairs Department emergency-relief page [7]. Law et al. (2026) was read as a neighbouring daily evaluation, not as this panel [9].

No new Hospital Authority model was fitted. No warning-day treatment indicator was constructed. Indoor temperature, air-conditioning use, and shelter occupancy were not measured.

What this mapping is not

This is not an evaluation of the Very Hot Weather Warning, the Cold Weather Warning, the Prolonged Heat Special Alert, the Hong Kong Heat Index, Labour heat-stress guidance, or a municipal HHAP. It does not count admissions averted. It does not set a five-day consecutive trigger. It does not treat Law et al. (2026) as a result of the CHD/HF file.

Results

WHO 2026 outputs

The 2026 guidance lists eight core elements: governance; a heat–health warning system; populations at increased risk; communication; health-system resilience; reducing heat exposure; heat–health surveillance; and monitoring, evaluation and learning [1]. The documented outputs that “collectively form a comprehensive HHAP” are:

1. an agreed governance structure with a lead body and a coordination mechanism;
2. an operational warning system that links each alert level to predefined measures, responsibilities, and timelines;
3. an assessment and mapping of populations at increased risk, with a specific outreach strategy;
4. a documented communication strategy;
5. a health-system resilience strategy with operational plans and business continuity arrangements;
6. a roadmap for heat-exposure reduction at household, building, and urban scales;
7. an operational heat–health surveillance system with indicators, data sources, reporting timelines, and a link to warnings;
8. a functional monitoring, evaluation and learning framework [1].

Warnings that are not linked to predefined actions are not, on this definition, a complete HHAP. Public advice that names a risk group is not, on this definition, a mapped outreach strategy. Monthly hospital counts are not operational surveillance for activation.

Hong Kong instruments against those outputs

Table 1 records the map. The last column is deliberately a refusal: the monthly panel cannot fill a missing WHO output.

Table 1. WHO 2026 core elements mapped onto named Hong Kong instruments.

Element	Named Hong Kong instrument	WHO 2026 output in hand?
1. Governance	Multi-agency practice: Observatory, Department of Health / Centre for Health Protection, Labour, Home Affairs	Related practice only. No verified HHAP governance document
2. Heat–health warning system	Very Hot Weather Warning; Hot Weather Special Advisory; Prolonged Heat Special Alert (16 July 2022); Extremely Hot Weather 35 °C or above; Hong Kong Heat Index; Labour heat-stress notes [3–5]	Operational products exist. A full alert-to-action package with assigned timelines was not verified
3. Populations at increased risk	Centre for Health Protection names infants, elderly people, pregnant women, people with heart disease or high blood pressure, outdoor workers, and overweight people [6]. The Observatory names elderly people, pregnant women, infants, children, and people with chronic illness [3]	Related naming. Not a mapped outreach strategy
4. Communication	Observatory website, app, and broadcast; Centre for Health Protection heat-stroke advice; Labour materials	Related messaging. Not a documented communication-strategy output
5. Health-system resilience	Hospital Authority operations exist and were not studied	Output not verified
6. Reducing heat exposure	Indoor ventilation and air-conditioning advice; temporary heat shelters [6,7]	Related advice and shelters. Not a household–building–urban roadmap
7. Heat–health surveillance	No near-real-time first-event CHD/HF feed	Absent for this estimand
8. Monitoring, evaluation and learning	Not performed here	Absent

What the monthly CHD/HF panel cannot do, by row: (1) judge leadership or financing; (2) evaluate whether any alert “works”, set a consecutive-day trigger, or test the hourly nighttime metric of Guo et al. [10]; (3) add a named group or target age, sex, or subtype (strata were not delivered); (4) test messages, behaviour change, or admissions averted; (5) audit staffing, ward cooling, or continuity; (6) measure indoor temperature, air-conditioning, or shelter use; (7) guide activation from monthly aggregates; (8) treat this paper as an evaluation.

The Observatory’s warning page states that cold and very hot weather warnings also alert departments such as the Home Affairs Department to consider opening temporary shelters [3]. That sentence is a link from warning to a possible action. It is not a published matrix of measures, owners, and start–stop rules for every alert level.

Chong and Law describe the Prolonged Heat Special Alert as applying when the Very Hot Weather Warning is in force and a few days of very hot days *or* hot nights are expected [4]. The service was announced on 16 July 2022 [5]. Hot nights are therefore already named beside very hot days in an operational product. That fact does not require, and is not tested by, a monthly count ratio of 1.022.

A neighbouring daily paper is a different estimand

Law, Chong, and colleagues analysed daily heat-related mortality and emergency calls in Hong Kong during 2012–2022 [9]. They used generalised linear models and a distributed-lag non-linear model on daily maximum temperature. They excluded 2019–2020 and tropical-cyclone or black-rainstorm days. Among people aged 0–64 years, the relative risk for heat-health mortality at a maximum temperature of 35.5 °C was 1.148 (0.832–1.584). They concluded that an advisory message at 35 °C was supportable. Daily minimum temperature showed elevated risk at extreme heat for the outcomes they studied.

That paper is daily, cause-grouped mortality and calls, and a maximum-temperature advisory question. The companion identification article is monthly first hospitalisation after first CHD or HF diagnosis in a diabetes or hypertension cohort, without admission cause, using official day counts divided by five. Extremely hot days (maximum temperature of 35 °C or above) were obtained with the weather file and were not a Model 1 exposure. The two papers must not be collapsed. Law et al. (2026) is not an evaluation of the Very Hot Weather Warning *by this panel*.

Cold weather is not a WHO heat element, and it is not optional here

WHO 2026 is heat guidance. Hong Kong also issues a Cold Weather Warning and opens temporary cold shelters [3,7]. In the companion identification article, the HF association with official cold days is the more

coherent of the two residual signals: 1.073 (1.006–1.144) in the full window and 1.113 (1.053–1.176) before 2020, concordant across four standard-error constructions, still unprotected by $q = 0.192$ [8]. A heat-only HHAP reading of that file would miss that residual. Mapping WHO heat elements onto Hong Kong does not retire the winter instrument.

What the monthly panel can be used for, and what it cannot

The 132-month panel can show that official cold days fall almost entirely in December–February, so the HF contrast is identified between winters [8]. It can show that a monthly official-day total does not distinguish consecutive days from days scattered through a month. It can refuse a recovered daily coefficient after a failed calibration. It cannot tell an on-call officer whether to activate a warning. It cannot estimate admissions averted. It cannot test indoor night temperature. It cannot replace Law et al.’s daily advisory question.

Table 2. What a warning evaluation would need, versus what the present files contain.

Need	This identification article	Law et al. (2026)	Still missing
Daily cause-recorded outcomes	Monthly first hospitalisation; no admission cause	Daily heat-related mortality and emergency calls	Cause-recorded cardiac admissions
Predeclared contrast	Twelve-contrast exploratory panel; all $q > 0.19$	Maximum-temperature advisory at 35 °C	Named alert, lag, population, multiplicity rule
Warning-day exposure	Official day counts / 5, not warning-on days	Daily maximum temperature, not the warning as treatment	Warning-on versus similar warning-off days
Surveillance linked to activation	Territory-month aggregates after the month closes	Not an operational feed	Element 7: indicators, sources, timelines
Learning cycle for thresholds	Not this paper	Not this paper	Element 8: discovery and confirmation separated

Table 2 is a design map. It is not a ranking of papers and it is not an evaluation of any warning.

Discussion

The 2026 guidance asks countries to produce eight documents, not eight slogans [1]. Hong Kong already has several of the *instruments* that those documents would organise: graded heat products, a night-aware prolonged-heat alert, named cardiac risk groups, and shelters. What is not in hand, in the sources reviewed here, is the HHAP as a single, evaluated, surveillance-linked plan. Saying that Hong Kong has no heat–health instruments is false. Saying that this research programme has evaluated an HHAP is also false.

Martinez et al. noted that warning systems can exist while evaluation and longer-term exposure reduction lag [2]. Hong Kong fits that pattern on the evidence assembled here: warnings and advice are public; a functional monitoring-evaluation-learning framework for those warnings is not this paper.

For a supervisor whose passion is HHAPs, the next scientific object is therefore not a larger coefficient in the identification article’s Table 2. It is an evaluation design that the present file cannot run:

1. Daily outcomes with recorded admission cause, or at least a principal diagnosis.
2. A predeclared contrast: which alert, which lag, which population, which multiplicity rule.
3. Warning-day exposure, with an explicit counterfactual for days that would have been similar without the alert.
4. Indicators, reporting timelines, and a link from surveillance to activation, as required for core element 7.
5. A learning cycle that can change a threshold without using the same data as both discovery and confirmation.

Until those exist, the HHAP contribution of this programme is the map in Table 1 and the design refusal in Table 2.

Physiology is a separate companion question. Sleep fragmentation, nocturnal core temperature in seven men, and bedroom heart-rate variability in older adults do not measure first CHD hospitalisation in Hong Kong and do not measure indoor temperature in this series. They are not used here as identified mechanisms.

Conclusion

Hong Kong already operates heat and cold instruments that overlap published HHAP elements. The 2026 guidance still asks for eight documented outputs. This paper maps the overlap and records the gaps. It does not evaluate warnings, does not count admissions averted, and does not promote a monthly count ratio to a

heat-plan finding. The companion identification article remains an identification article. Table 2 states the evaluation that is still missing. Better evaluation will need daily, cause-recorded, warning-day data.

Strengths and limitations

Strengths. Each row of Table 1 names a primary source. WHO elements are copied from the 2026 output list rather than paraphrased into policy slogans. The monthly CHD/HF panel is kept in its own estimand. Law et al. (2026) is cited as a neighbour, not imported.

Limitations. We did not obtain an official statement that Hong Kong has, or has not, adopted a document titled heat-health action plan. Absence of a WHO output in the public pages reviewed is not proof that an internal plan does not exist. Labour heat-stress rules and Hospital Authority continuity plans were not audited as documents. Chinese-language service pages were not treated as a separate corpus. The mapping cannot be stronger than the public instruments.

Data and code

Code for the companion identification article is at <https://github.com/bobshenruililin/Laidlaw-Heat-Project>. Monthly hospital counts are not posted. This mapping uses only public WHO, Observatory, Department of Health, and Home Affairs pages plus published papers.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgements

None.

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